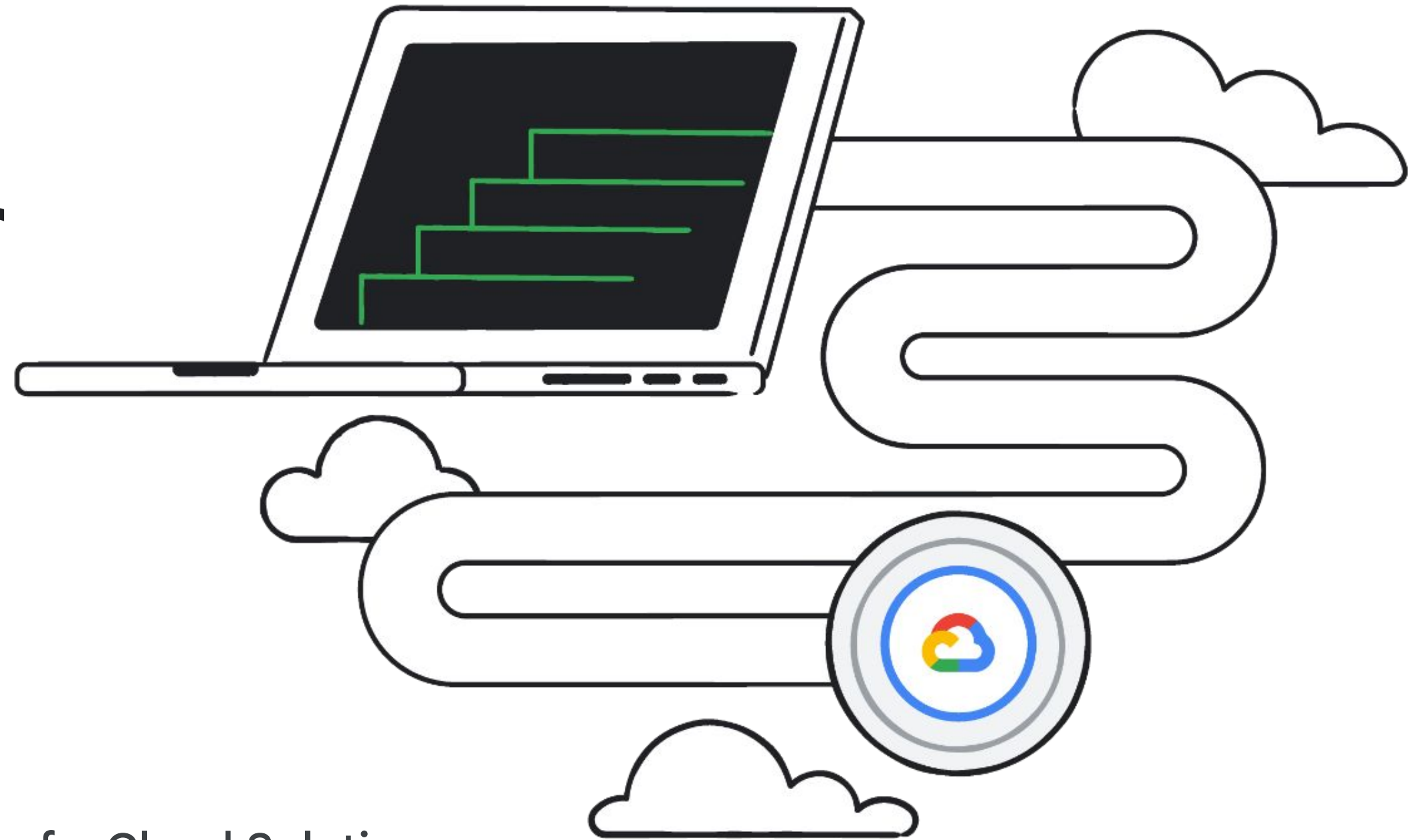
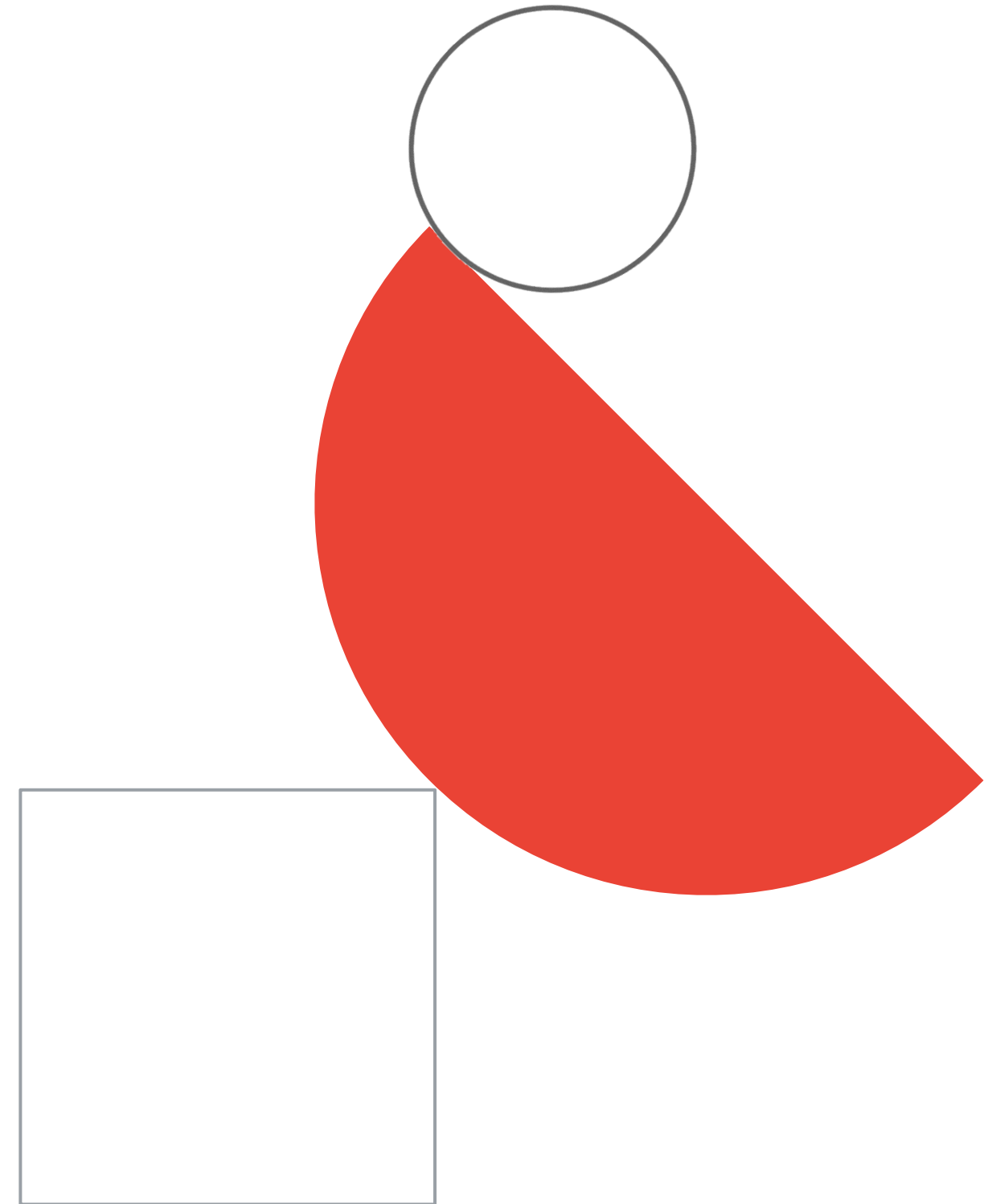


Preparing for Your Associate Cloud Engineer Journey



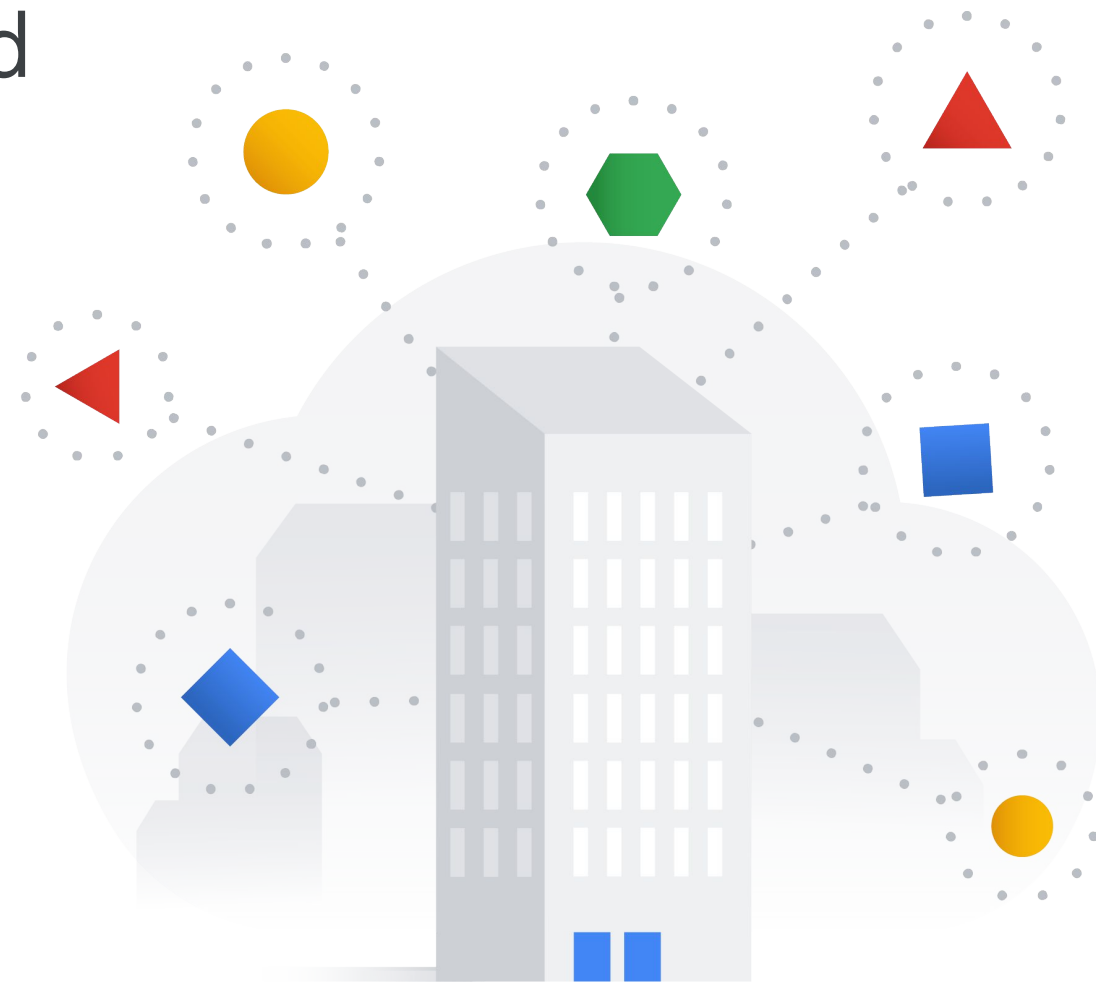
Module 4: Ensuring Successful Operation of a Cloud Solution

Managing Cymbal Superstore's cloud solutions



In this course:

managing Cymbal
Superstore's cloud
solutions



- Managing Compute Engine resources
- Managing GKE resources
- Managing Cloud Run resources
- Managing storage and database solutions
- Managing networking resources
- Monitoring and logging



4.1 | Managing Compute Engine resources

Considerations include:

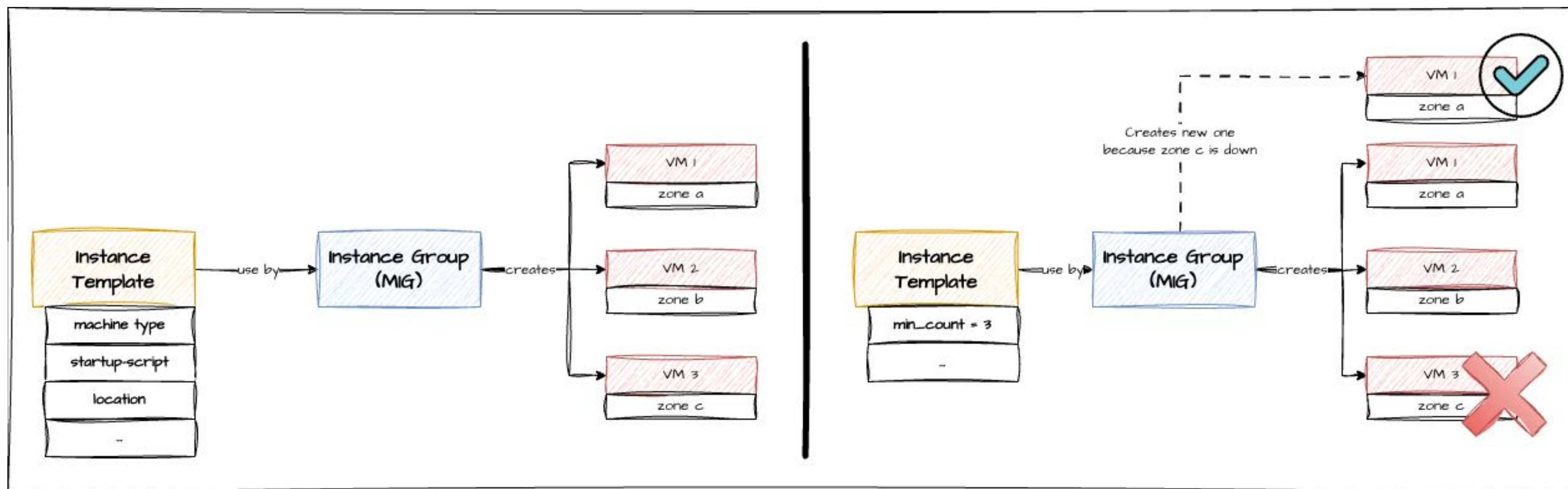
- **Managed instance groups**
- **Viewing current running VM inventory (e.g., instance IDs, details)**
- **Remotely connecting to the instance**
- Working with snapshots (e.g., create a snapshot from a VM, view snapshots, delete a snapshot, schedule a snapshot)
- Working with images (e.g., create an image from a VM or a snapshot, view images, delete an image)

Managing Cymbal Superstore's **supply chain app**

Example for upgrading **managed instance groups** based on a template:

```
gcloud compute instance-groups managed  
rolling-action start_update cymball_supplychain_ig \  
  --version=template=cymball_supplychain_ig_templat  
e_<yymmdd> \  
  --type=proactive\  
  --region=us-central1
```


What are templates and managed instance groups?



4.1 | Managing Compute Engine resources

Considerations include:

- Managed instance groups
- Viewing current running VM inventory (e.g., instance IDs, details)
- Remotely connecting to the instance
- **Working with snapshots (e.g., create a snapshot from a VM, view snapshots, delete a snapshot, schedule a snapshot)**
- Working with images (e.g., create an image from a VM or a snapshot, view images, delete an image)

4.1 | Diagnostic Question 01 Discussion



You want to view a description of your available snapshots using the command line interface (CLI). What gcloud command should you use?

- A. gcloud compute snapshots list
- B. gcloud snapshots list
- C. gcloud compute snapshots get
- D. gcloud compute list snapshots

4.1 | Diagnostic Question 01 Discussion



You want to view a description of your available snapshots using the command line interface (CLI). What gcloud command should you use?

- A. **gcloud compute snapshots list**
- B. gcloud snapshots list
- C. gcloud compute snapshots get
- D. gcloud compute list snapshots



Getting information about snapshots

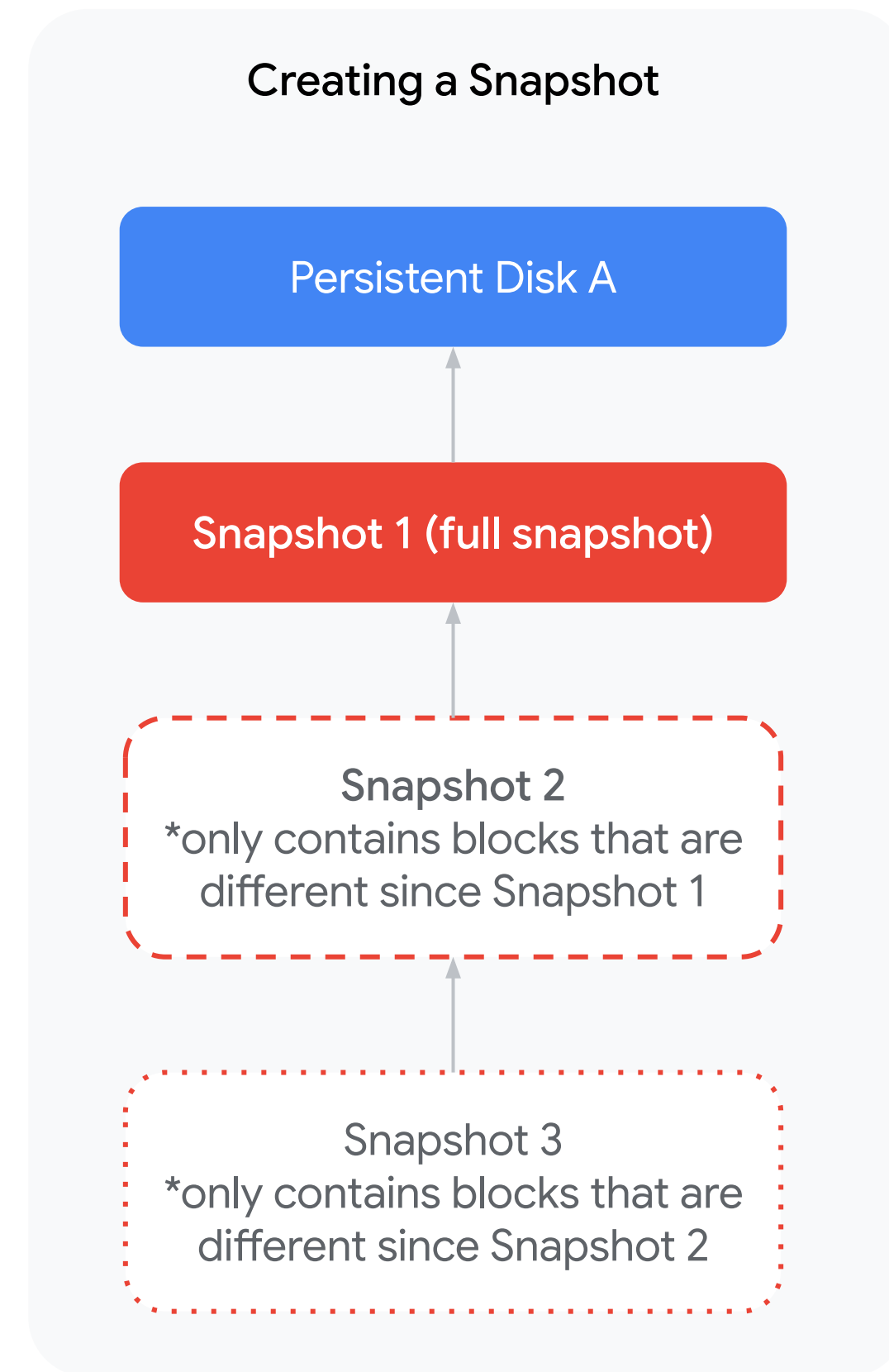
To list Compute Engine disk snapshots:

```
gcloud compute snapshots list --project PROJECT_ID
```

To describe snapshots:

```
gcloud compute snapshots describe SNAPSHOT_NAME
```

Snapshots are
incremental



4.1 | Managing Compute Engine resources

Considerations include:

- Managed instance groups
- Viewing current running VM inventory (e.g., instance IDs, details)
- Remotely connecting to the instance
- Working with snapshots (e.g., create a snapshot from a VM, view snapshots, delete a snapshot, schedule a snapshot)
- **Working with images (e.g., create an image from a VM or a snapshot, view images, delete an image)**

Images: Blueprints for Boot Disks

An image contains:

- The **Operation System** (e.g. Debian, Windows Server, etc.)
- The **Boot Loader** (to start the kernel)
- Preinstalled **Software and Libraries** (e.g. Python, NgINX, daemons, etc.)
- **File System Structure**

Images are:

- **Immutable disk structure** to create the boot disk

4.1

Managing Compute Engine resources

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M3 Virtual Machines and Networks in the Cloud

[Architecting with Google Compute Engine](#)

- M3 Virtual Machines
- M9 Load Balancing and Autoscaling



=

[Essential Google Cloud Infrastructure: Foundation](#)

- M3 Virtual Machines

[Elastic Google Cloud Infrastructure: Scaling and Automation](#)

- M2 Load Balancing and Autoscaling



Documentation

[Working with persistent disk snapshots | Compute Engine Documentation](#)

[Working with persistent disk snapshots | Compute Engine Documentation](#)

[Persistent disk snapshots | Compute Engine Documentation](#)

4.2 | Managing Google Kubernetes Engine resources

Considerations include:

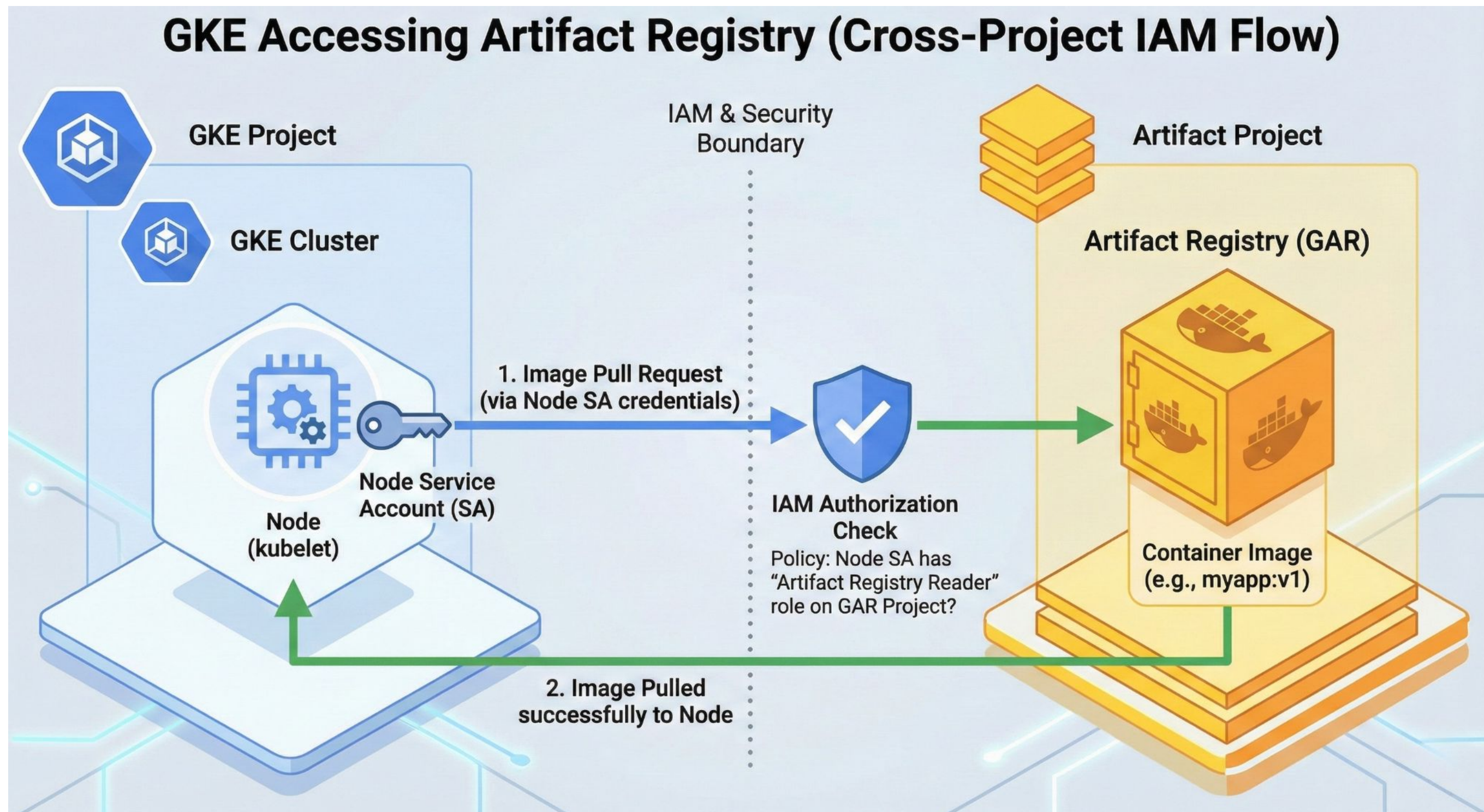
- **Viewing current running cluster inventory (e.g., nodes, pods, services)**
- Configuring Google Kubernetes Engine to access Artifact Registry
- Working with node pools (e.g., add, edit, or remove a node pool)
- Working with Kubernetes resources (e.g., Pods, Services, Statefulsets)
- Managing Horizontal and Vertical autoscaling configurations
- Load Balancing inside of GKE

4.2 | Managing Google Kubernetes Engine resources

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Configuring Google Kubernetes Engine to access Artifact Registry



4.2 | Managing Google Kubernetes Engine resources

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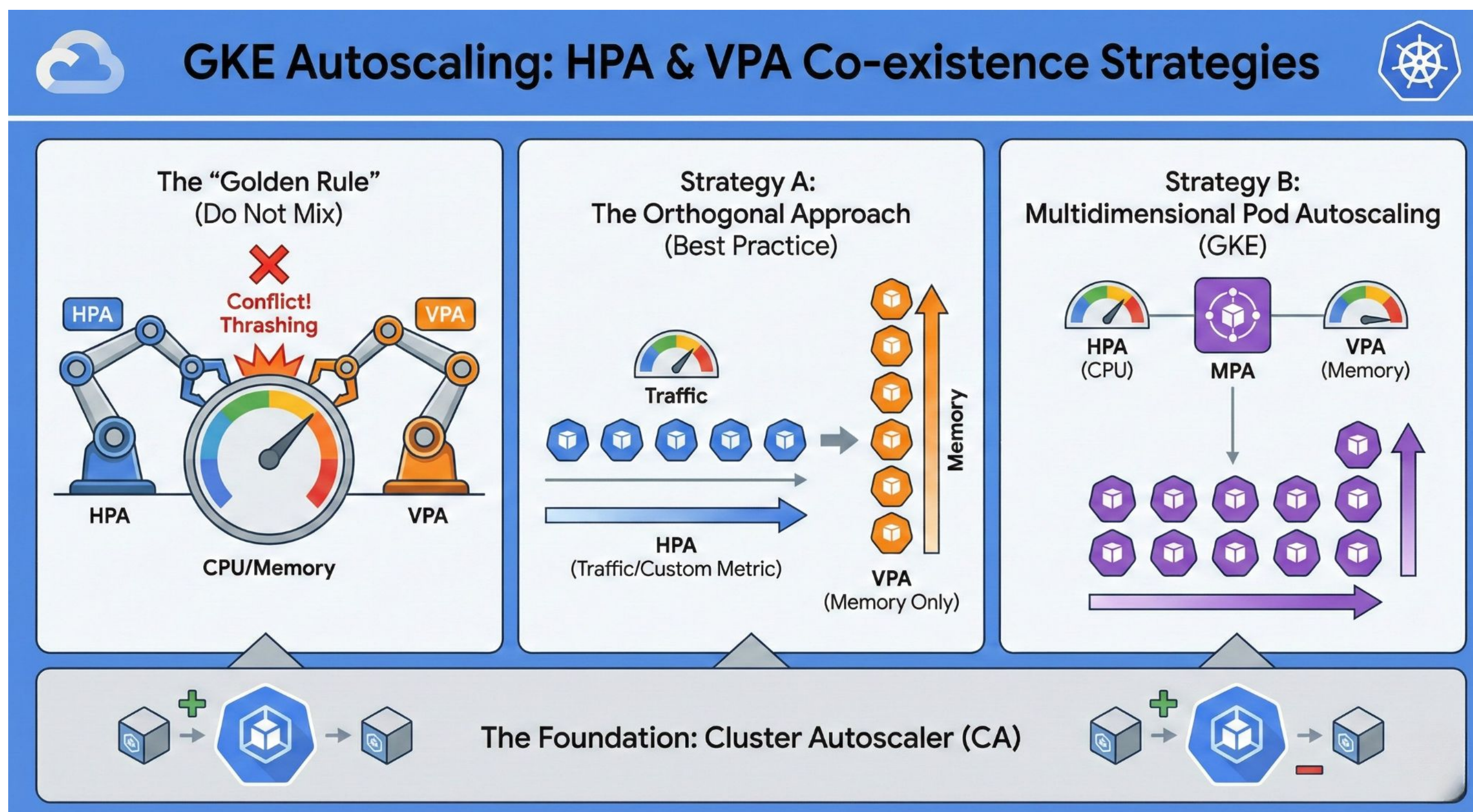
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- Load Balancing inside of GKE

Managing Horizontal and Vertical autoscaling configurations

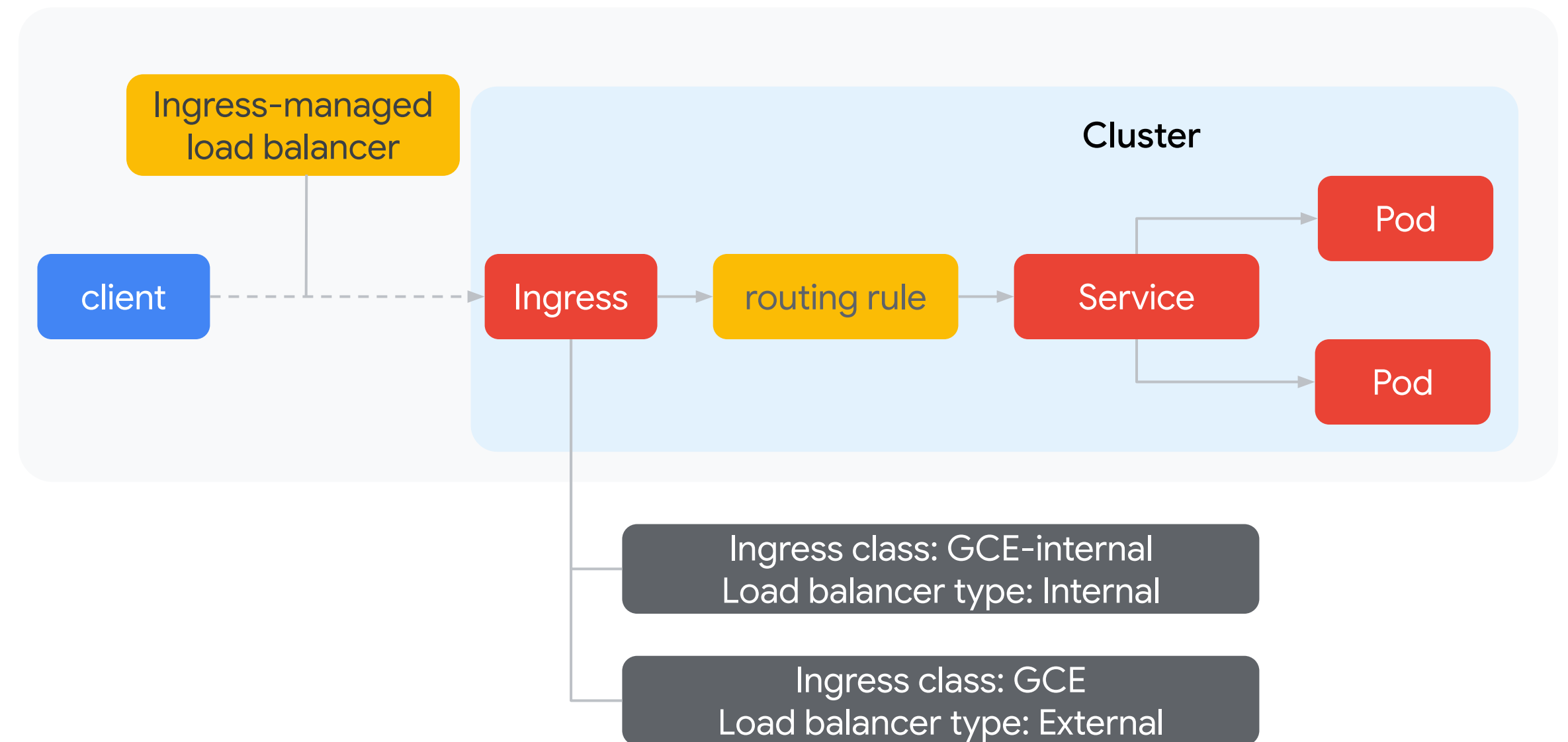


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- Viewing current running cluster inventory (e.g., nodes, pods, services)
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- Working with node pools (e.g., add, edit, or remove a node pool)
- Working with Kubernetes resources (e.g., Pods, Services, Statefulsets)
- Managing Horizontal and Vertical autoscaling configurations
- **Load Balancing inside of GKE**

Internal vs External load balancing in Kubernetes



4.2 | Diagnostic Question 03 Discussion



Cymbal Superstore's GKE cluster requires an internal Application Load Balancer. You are creating the configuration files required for this resource.

What is the proper setting for this scenario?

- A. Annotate your ingress object with an `ingress.class` of "gce."
- B. Configure your service object with a type: `LoadBalancer`.
- C. Annotate your service object with a "neg" reference.
- D. Implement custom static routes in your VPC.

4.2 | Diagnostic Question 03 Discussion



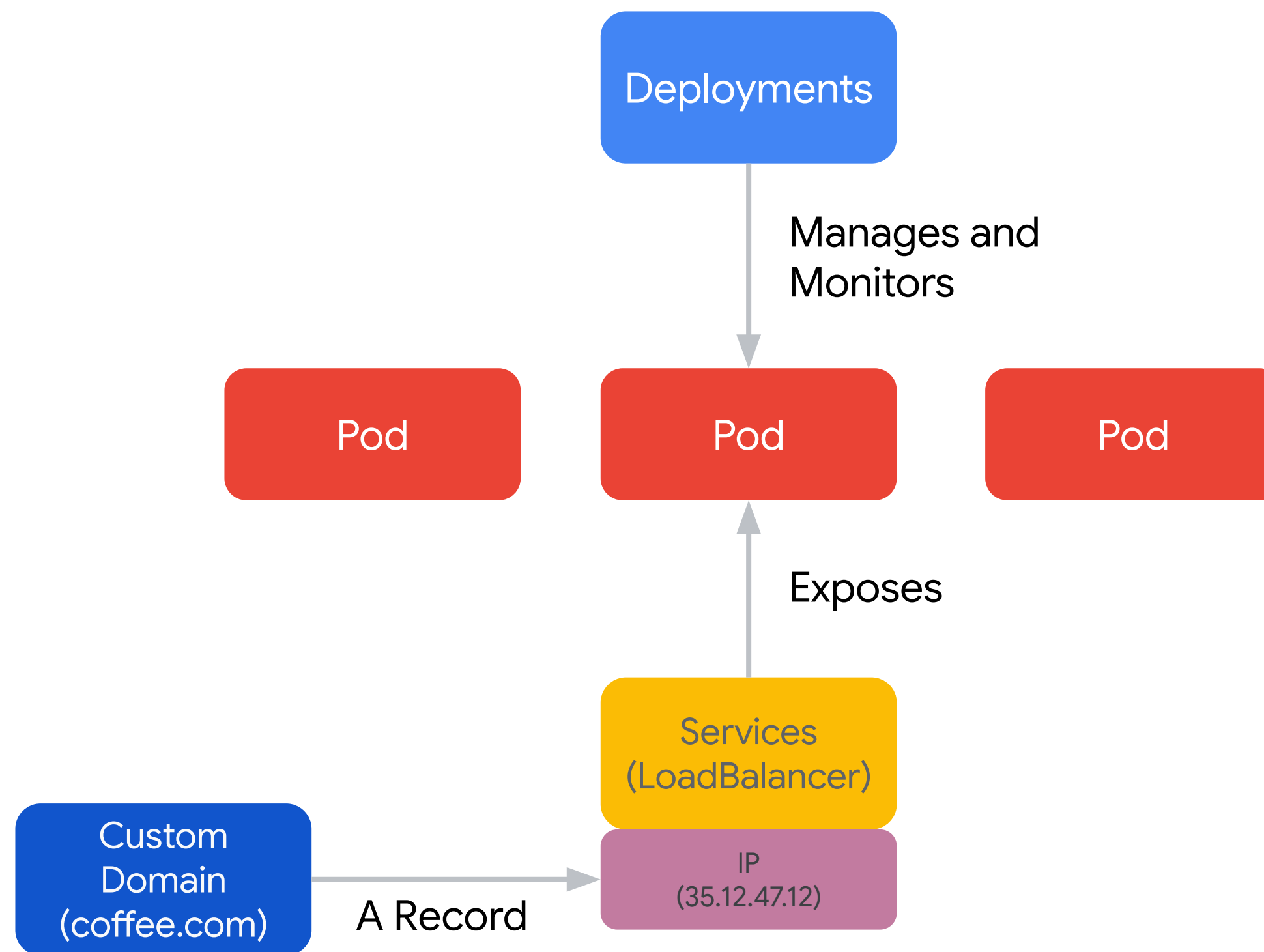
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- B. Configure your service object with a type: `LoadBalancer`.
- C. Annotate your service object with a "neg" reference.
- D. Implement custom static routes in your VPC.



Kubernetes objects



4.2 | Diagnostic Question 04 Discussion



What Kubernetes object provides access to logic running in your cluster via endpoints that you define?

- A. Pod templates
- B. Pods
- C. Services
- D. Deployments

4.2 | Diagnostic Question 04 Discussion

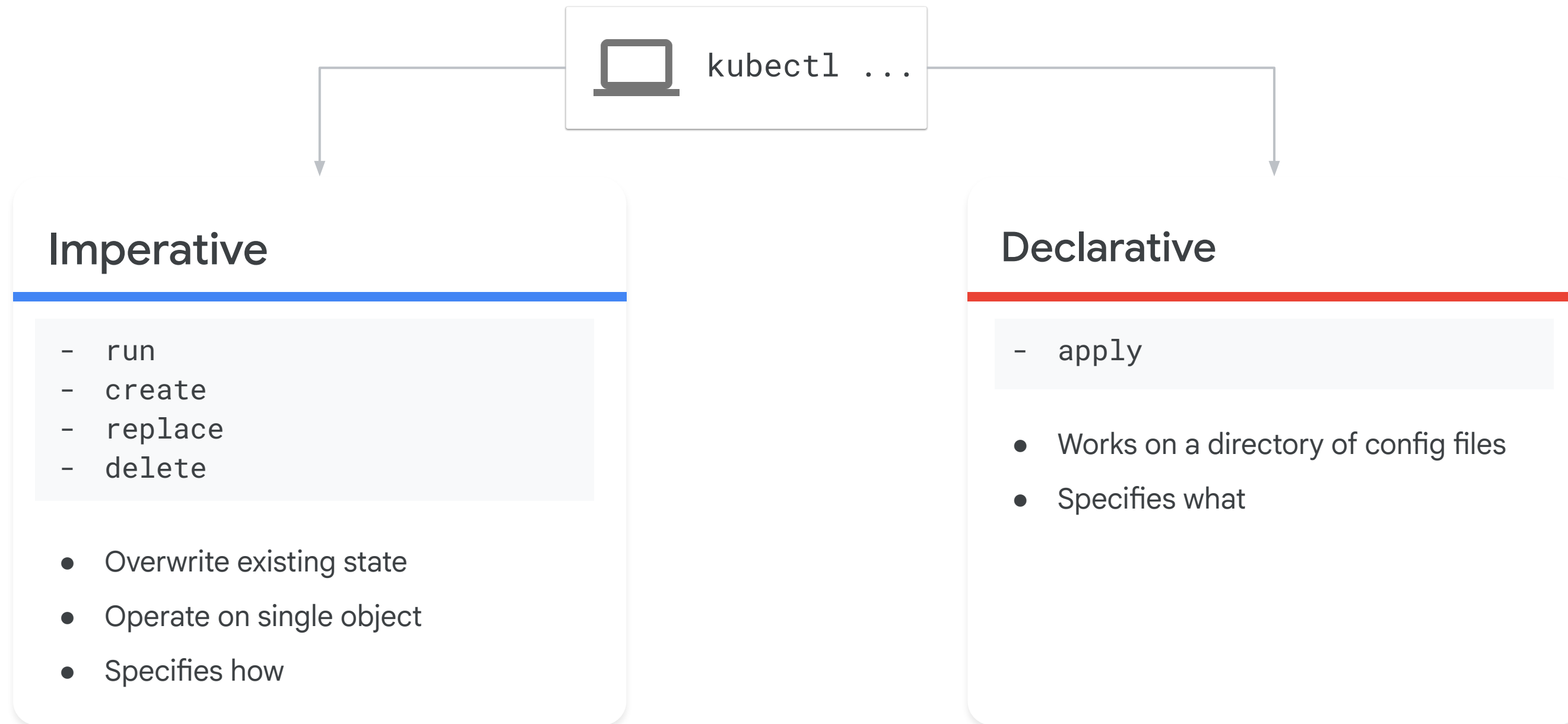


What Kubernetes object provides access to logic running in your cluster via endpoints that you define?

- A. Pod templates
- B. Pods
- C. Services**
- D. Deployments



Types of kubectl commands



4.2 | Diagnostic Question 05 Discussion

What is the declarative way to initialize and update Kubernetes objects?

- A. kubectl apply
- B. kubectl create
- C. kubectl replace
- D. kubectl run



4.2 | Diagnostic Question 05 Discussion



What is the declarative way to initialize and update Kubernetes objects?

- A. `kubectl apply`
- B. `kubectl create`
- C. `kubectl replace`
- D. `kubectl run`



4.2

Managing Google Kubernetes Engine resources

Courses

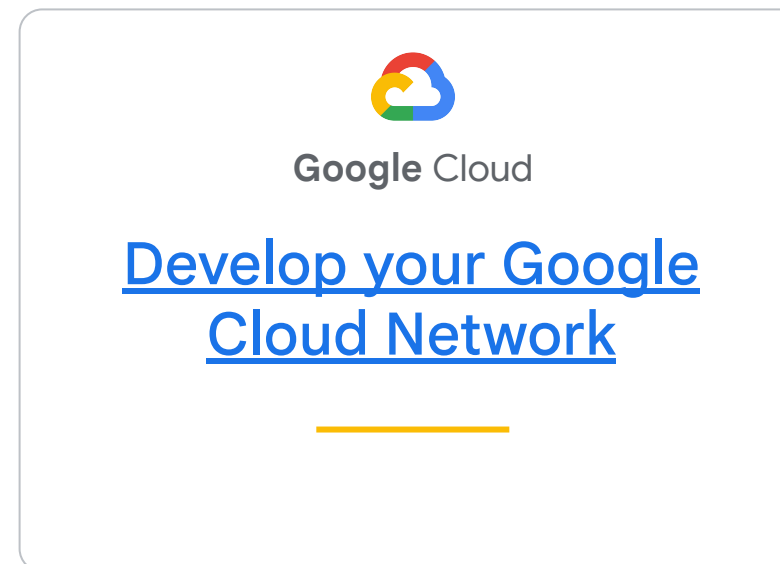
[Google Cloud Fundamentals: Core Infrastructure](#)

- M5 Containers in the Cloud

[Getting Started with Google Kubernetes Engine](#)

- M3 Kubernetes Architecture
- M4 Kubernetes Operations

Skill Badge



Documentation

[Ingress for internal Application Load Balancers](#)

[Ingress for external Application Load Balancers](#)

[Configure Ingress for external Application Load Balancers](#)

[Configuring Ingress for internal Application Load Balancers](#)

[GKE overview | Kubernetes Engine Documentation](#)

[Pod | Kubernetes Engine Documentation](#)

[Deployment | Kubernetes Engine Documentation](#)

[Services | Kubernetes Engine Documentation](#)

[Overview of deploying workloads | Kubernetes Engine Documentation](#)

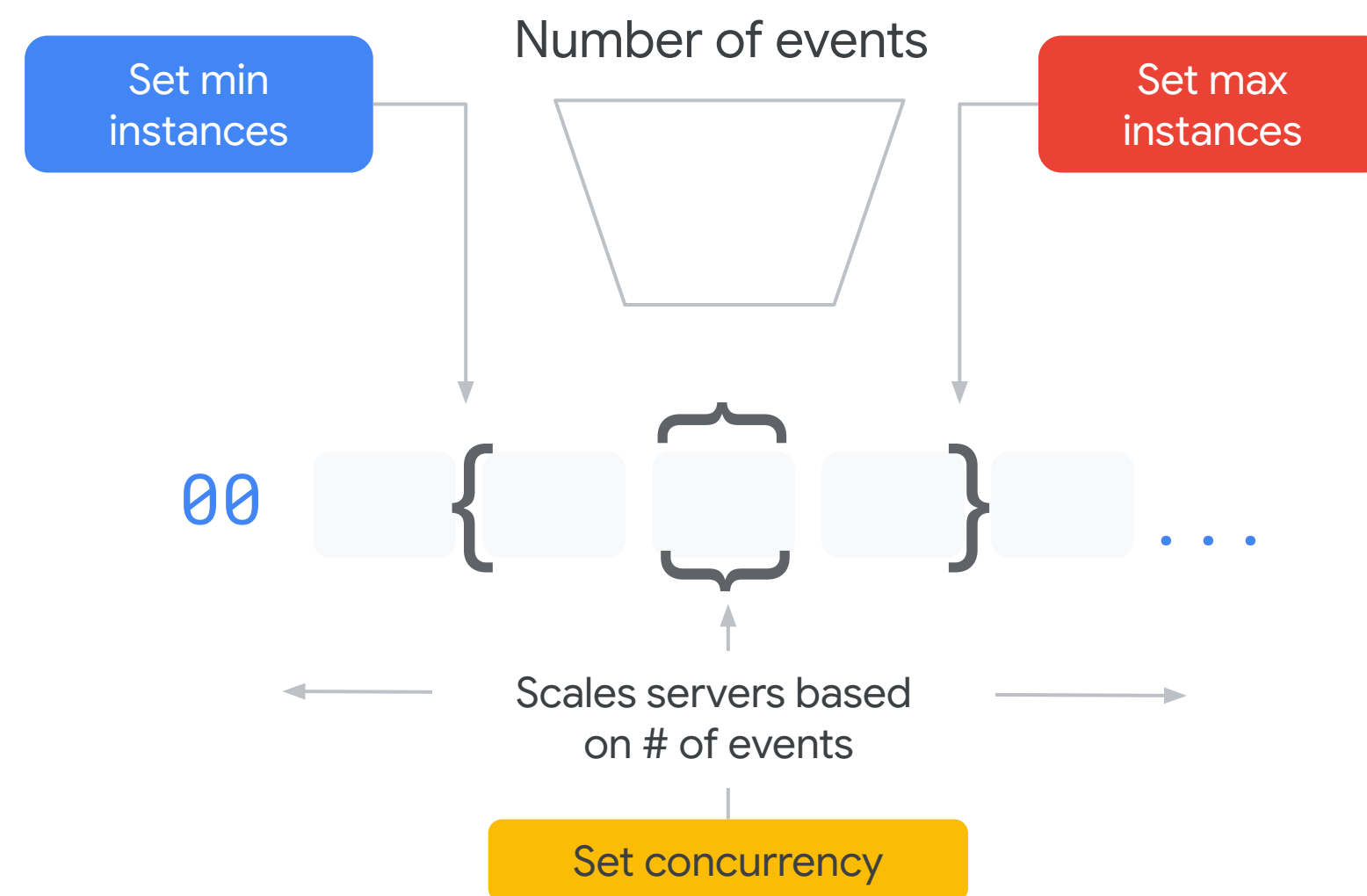
[Kubernetes Object Management](#)

4.3 | Managing Cloud Run resources

Considerations include:

- Deploying new versions of an application
- Adjusting application traffic splitting parameters
- Setting scaling parameters for autoscaling instances

Cloud Run autoscaling



4.3 | Diagnostic Question 06 Discussion



You have a Cloud Run service with a database backend. You want to limit the number of connections to your database.

- A. Set Min instances.
- B. Set Max instances.
- C. Set CPU Utilization.
- D. Set Concurrency settings.

What should you do?

4.3 | Diagnostic Question 06 Discussion



You have a Cloud Run service with a database backend. You want to limit the number of connections to your database.

What should you do?

- A. Set Min instances.
- B. Set Max instances.**
- C. Set CPU Utilization.
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4.3 | Managing Cloud Run resources

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M6 Applications in the Cloud

Documentation

[About container instance autoscaling | Cloud Run Documentation](#)

4.4 | Managing storage and database solutions

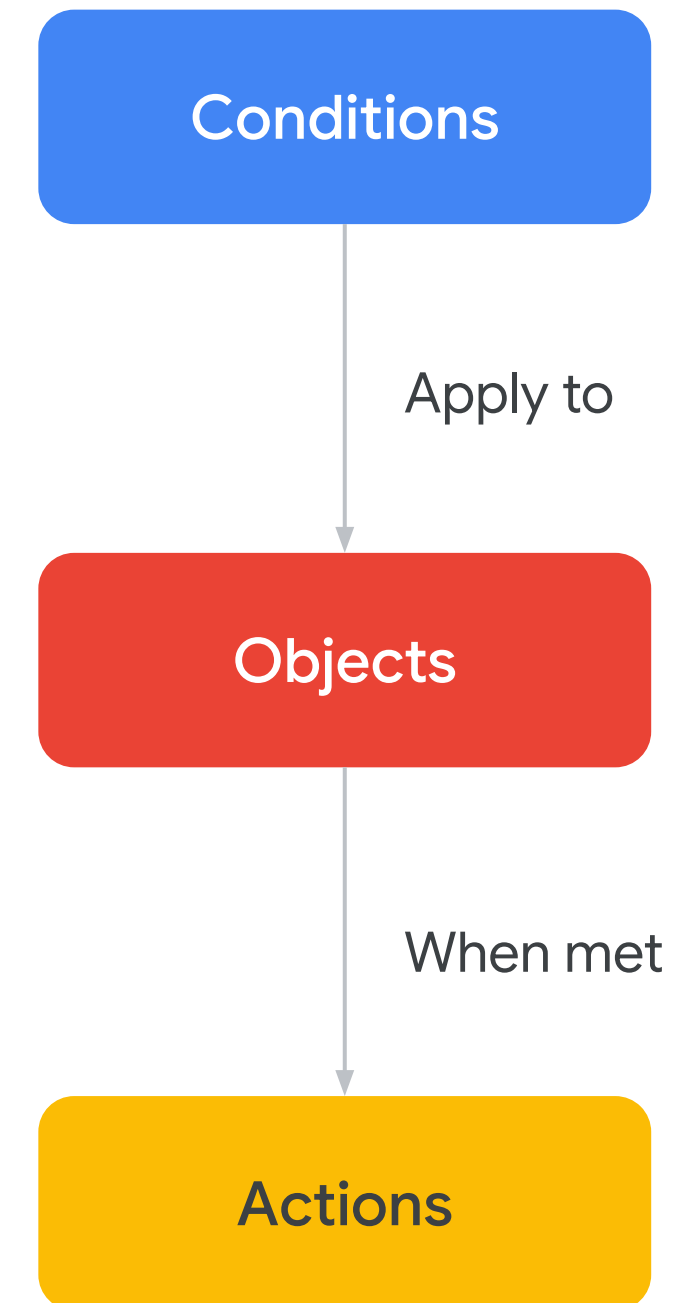
Considerations include:

- Managing and securing objects in Cloud Storage buckets
- Setting object lifecycle management policies for Cloud Storage buckets
- Executing queries to retrieve data from data instances (e.g., Cloud SQL, BigQuery, Spanner, Firestore, AlloyDB)
- Estimating costs of data storage resources
- Backing up and restoring database instances (e.g., Cloud SQL, Firestore)
- Reviewing job status (e.g., Dataflow, BigQuery)

Cloud Storage Lifecycle Actions

- Age
- Createdbefore
- Customtimebefore
- ...

- Delete
- SetStorageClass
-



4.4 | Diagnostic Question 07 Discussion



You want to implement a lifecycle rule that changes your storage type from Standard to Nearline after a specific date.

What conditions should you use?
(Pick two.)

- A. Age
- B. CreatedBefore
- C. MatchesStorageClass
- D. IsLive
- E. NumberofNewerVersions

4.4 | Diagnostic Question 07 Discussion



You want to implement a lifecycle rule that changes your storage type from Standard to Nearline after a specific date.

What conditions should you use?
(Pick two.)

- A. Age
- B. CreatedBefore**
- C. MatchesStorageClass**
- D. IsLive
- E. NumberofNewerVersions



4.4 | Managing storage and database solutions

Courses

[Google Cloud Fundamentals: Core Infrastructure](#)

- M4 Storage in the Cloud

[Architecting with Google Compute Engine](#)

- M5 Storage and Database Services



=

[Essential Google Cloud Infrastructure: Core Services](#)

- M2 Storage and Database Services



Documentation

[Object Lifecycle Management | Cloud Storage](#)

4.5 | Managing networking resources

Considerations include:

- Adding a subnet to an existing VPC
- Expanding a subnet to have more IP addresses
- Reserving static external or internal IP addresses
- Working with Cloud DNS and Cloud NAT

Expand IP addresses in a subnet

Current IP address range

Reduce your mask:
e.g. 24 to 20

Expanded address range

4.5 | Diagnostic Question 08 Discussion



Cymbal Superstore has a subnetwork called mysubnet with an IP range of 10.1.2.0/24. You need to expand this subnet to include enough IP addresses for at most 2000 users or devices.

What should you do?

- A. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 20`
- B. `gcloud networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21`
- C. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 21`
- D. `gcloud compute networks subnets expand-ip-range mysubnet --region us-cetnral1 --prefix-length 22`

4.5 | Diagnostic Question 08 Discussion



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- D. `gcloud compute networks subnets expand-ip-range mysubnet --region us-central1 --prefix-length 22`



4.5 | Managing networking resources

Courses

[Architecting with Google Compute Engine](#)

- M2 Virtual Networks



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[Essential Google Cloud Infrastructure: Foundation](#)

- M2 Virtual Networks



Documentation

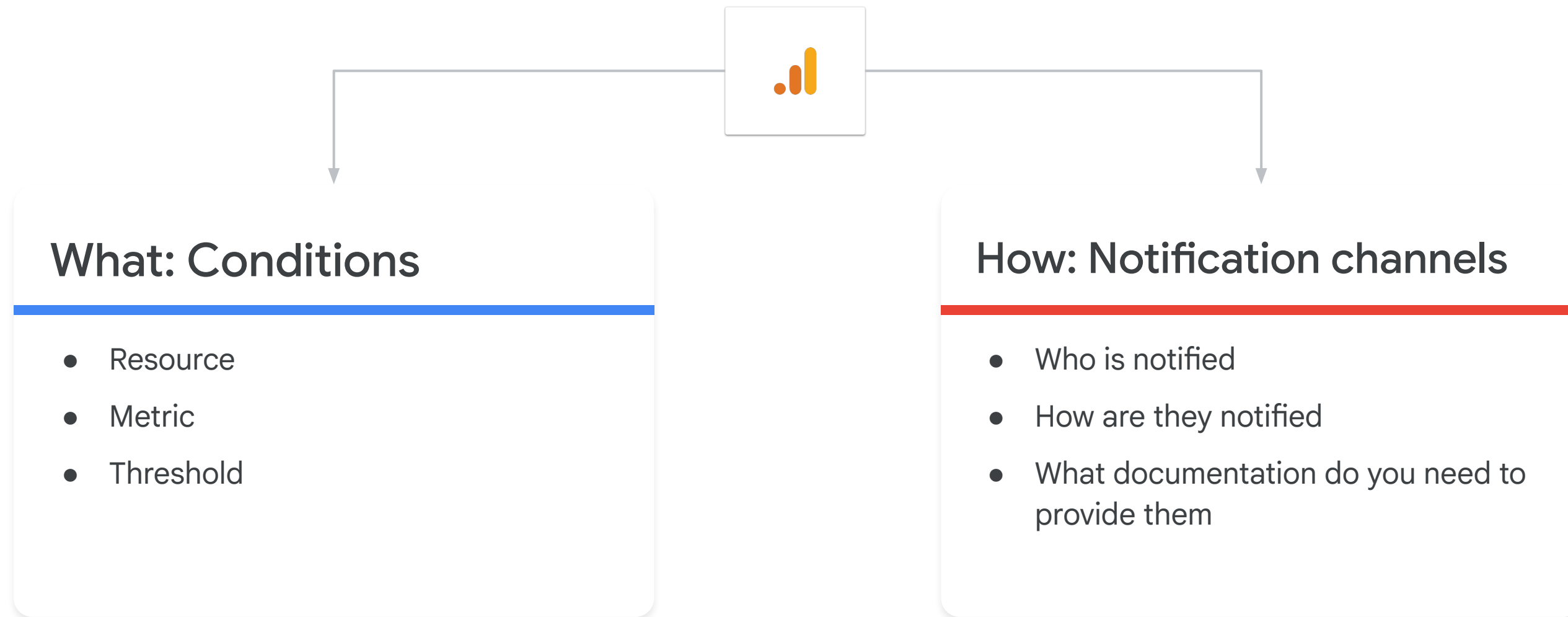
[gcloud compute networks subnets expand-ip-range](#)
[Using VPC networks](#)

4.6 | Monitoring and logging

Considerations include:

- Creating Cloud Monitoring alerts based on resource metrics
- Creating and ingesting Cloud Monitoring custom metrics (e.g., from applications or logs)
- Exporting logs to external systems (e.g., on-premises, BigQuery)
- Configuring log buckets, log analytics, and logs routers
- Viewing and filtering logs in Cloud Logging
- Viewing specific log message details in Cloud Logging
- Using cloud diagnostics to research an application issue
- Viewing Google Cloud status
- **Configuring and deploying Ops Agent**
- Deploying Managed Service for Prometheus (GKE)
- Configuring audit logs

Cloud operations custom alerts



4.6 | Diagnostic Question 09 Discussion

Cymbal Superstore's supply chain management system has been deployed and is working well. You are tasked with monitoring the system's resources so you can react quickly to any problems. You want to ensure the CPU usage of each of your Compute Engine instances in us-central1 remains below 60%. You want an incident created if it exceeds this value for 5 minutes. You need to configure the proper alerting policy for this scenario.

What should you do?

- A. Choose resource type of VM instance and metric of CPU load, condition trigger if any time series violates, condition is below, threshold is .60, for 5 minutes.
- B. Choose resource type of VM instance and metric of CPU utilization, condition trigger all time series violates, condition is above, threshold is .60 for 5 minutes.
- C. Choose resource type of VM instance, and metric of CPU utilization, condition trigger if any time series violates, condition is below, threshold is .60 for 5 minutes.
- D. Choose resource type of VM instance and metric of CPU utilization, condition trigger if any time series violates, condition is above, threshold is .60 for 5 minutes.



4.6 | Diagnostic Question 09 Discussion

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4.6 | Monitoring and logging

Courses

[Architecting with Google Compute Engine](#)

- M7 Resource Monitoring



=

[Essential Google Cloud Infrastructure: Core Services](#)

- M4 Resource Monitoring



Skill Badges



Google Cloud

[Set Up an App Dev Environment on Google Cloud](#)



Google Cloud

[Develop your Google Cloud Network](#)

Documentation

[Managing metric-based alerting policies | Cloud Monitoring](#)

[Introduction to alerting | Cloud Monitoring](#)

Knowledge Check 1

What GKE object implements an Application Load Balancer?

- A. Service
- B. Pod
- C. Deployment
- D. Ingress



Knowledge Check 1

What GKE object implements an Application Load Balancer?

- A. Service
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Knowledge Check 2

Which Cloud Run autoscaling setting should you set if you want to limit cost?

- A. Min Instances
- B. Max instances
- C. Concurrency settings
- D. CPU utilization



Knowledge Check 2

Which Cloud Run autoscaling setting should you set if you want to limit cost?

A. Min Instances

B. Max instances

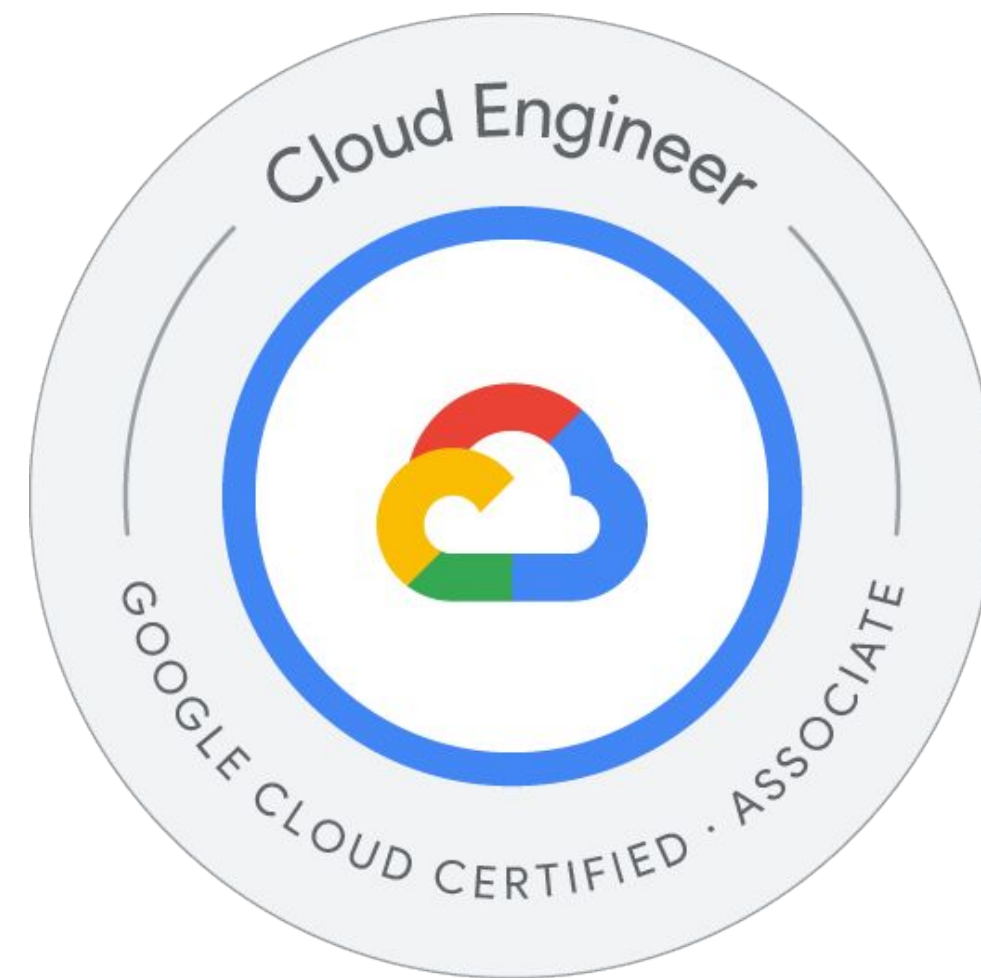
C. Concurrency settings

D. CPU utilization



Make sure to...

Enjoy the journey as much
as the destination!



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